

SMARTBRIDGE / SKILL WALLET

Functional Features

Date	2 July 2026	Team ID	—
Project Name	CaptionAI: AI-Powered Social Media Caption Generator		

Functional Features Included in the Solution

Provide a comprehensive list of all functional features that have been successfully integrated into your final solution. This helps evaluators understand the scope of your project, the modules you have developed, and the exact capabilities your application provides to the end users.

SN	Feature Name	Description & Purpose	Module / Layer	Status
1	Caption Generator Layer	Produces three unique, platform-optimized captions using the Groq API (Llama-3.3-70B-versatile). Boilerplate structures engineered via Gemini 1.5 Pro.	Backend LLM Engine	Completed
2	Hashtag Suggester Engine	Curates a strategic, balanced mix of 10 highly relevant viral and niche hashtags to optimize organic post discoverability.	Processing Core	Completed
3	Call-to-Action Utility	Automatically generates 3 punchy, action-oriented structural phrases targeted at boosting audience conversation and conversion metrics.	Engagement Layer	Completed
4	Asynchronous Control Portal	Single-page application dashboard implementing asynchronous background processing using AJAX Fetch requests without document reloads. Layout built via v0 by Vercel.	Frontend UI	Completed
5	Input Validation Guard	Securely filters and sanitizes user input parameters using custom regular expression logic while strictly enforcing a 2000-character ceiling.	Input Security Layer	Completed

SN	Feature Name	Description & Purpose	Module / Layer	Status
6	Isolated Credential Engine	Manages core application access tokens inside dedicated local configuration files (.env), separating private keys entirely from plain-text scripts.	<i>Security Config</i>	Completed
7	Secure Proxy Public Gateway	Establishes an explicit public-facing network interface tunnel on port 5000 using Ngrok to route external traffic securely to the local Python Flask engine.	<i>Network Infra</i>	Completed
8	One-Click Clipboard Core	Integrates an instantaneous text compilation and copying script with responsive, dynamic UI interface transitions providing visual feedback.	<i>Frontend Utility</i>	Completed